

# NLU course project

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The following is the report for the second part of the project:  
Natural Language Understanding.

## 1. Introduction

This part of the project covers two ATIS tasks trained jointly: slot filling, a sequence-labeling problem evaluated with CoNLL F1, and intent classification, evaluated with accuracy. Part A starts from a unidirectional LSTM joint model (ModelIAS) and incrementally adds bidirectionality and dropout, followed by a two-round sequential hyperparameter search. Part B instead fine-tunes a pre-trained BERT-base-uncased encoder for the same two tasks, following Chen et al. (2019), reusing as much of Part A's training/evaluation infrastructure as the different backbone allows. Both parts rank experiments by the average of Slot F1 and Intent Accuracy, alongside the two required metrics individually.

## 2. Implementation details

### 2.1. Part A

I implemented ModelIAS as the Baseline: a unidirectional LSTM encoder shared between a per-token slot-tagging head and an intent-classification head, with no dropout applied. Starting from this Baseline, I added the two required modifications one at a time. For bidirectionality, enabling the LSTM's `bidirectional` flag [1] required concatenating the forward and backward final hidden states before feeding the intent head. For dropout [2], I applied it after the embedding lookup and before both output heads. The two-round hyperparameter search that followed tuned learning rate, hidden/embedding size, dropout rate, batch size, and gradient clipping sequentially, ranking trials by the average of Slot F1 and Intent Accuracy rather than F1 alone, since the task is explicitly joint.

### 2.2. Part B

Part B fine-tunes BERT-base-uncased [3, 4] for the same joint task, reusing Part A's training loop structure where the different backbone allows. Following Chen et al. [5], sub-word tokenization is handled by assigning each word's slot label only to its first WordPiece sub-token, with an ignore index on every other position, so cross-entropy and evaluation are both driven only by first-sub-token positions; the [CLS] token's pooled output feeds the intent head directly, requiring no length-based sequence sorting since attention, unlike the LSTM, is not order-dependent. For the baseline, I matched the paper's reported `lr=5e-5`, `batch_size=128`, and `dropout=0.1`, but deliberately kept AdamW rather than switching to the paper's plain Adam, then ran the same averaged-metric sequential search used in Part A.

## 3. Results

### 3.1. Part A

Bidirectionality gave the largest single gain on its own, letting the slot head use both left and right context per token and giving the intent head a fuller summary of the whole utterance. Dropout added a further improvement on top of that. The two-round sequential search then settled on `lr=0.001`, `hidden_size=200`, `emb_size=200`, `dropout_rate=0.7`, `batch_size=64`, and `clip=0.25`, reaching a final Test F1 of 0.9513 and Intent Accuracy of 0.9653.

Table 1: Best results for each configuration (Part A).

| Configuration                         | Test F1       | Test Intent Acc |
|---------------------------------------|---------------|-----------------|
| Baseline (unidirectional, no dropout) | 0.9186        | 0.9339          |
| + Bidirectional                       | 0.9375        | 0.9362          |
| + Dropout                             | 0.9431        | 0.9518          |
| <b>Final (tuned)</b>                  | <b>0.9513</b> | <b>0.9653</b>   |

### 3.2. Part B

The BERT baseline, matching the paper's reported learning rate, batch size, and dropout (but keeping AdamW instead of the paper's plain Adam), already lands close to its published ATIS numbers. The subsequent search found a different optimum for this codebase and split: a smaller learning rate, a much smaller batch size, and a nonzero weight decay the paper never used, together reaching a final Test F1 of 0.9599 and Intent Accuracy of 0.9742 — reproduced exactly when retraining from scratch with the tuned configuration.

Table 2: Best results for each configuration (Part B).

| Configuration        | Test F1       | Test Intent Acc |
|----------------------|---------------|-----------------|
| Baseline (AdamW)     | 0.9569        | 0.9742          |
| <b>Final (tuned)</b> | <b>0.9599</b> | <b>0.9742</b>   |

Tuning was split across three rounds; the table below gives the best value found for each parameter, together with the average metric at the point it was fixed.

Table 3: Sequential search results across all rounds (Part B).

| Parameter    | Best Value | Best Avg |
|--------------|------------|----------|
| lr           | 3e-05      | 0.9671   |
| weight_decay | 0.1        | 0.9663   |
| dropout_rate | 0.1        | 0.9671   |
| batch_size   | 32         | 0.9671   |
| clip         | 1.0        | 0.9671   |
| n_epochs     | 20         | 0.9671   |

## 4. References

- [1] PyTorch, “Long short-term memory (lstm),” <https://docs.pytorch.org/docs/stable/generated/torch.nn.LSTM.html>.
- [2] —, “Dropout,” <https://docs.pytorch.org/docs/stable/generated/torch.nn.Dropout.html>.
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2019.
- [4] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020, pp. 38–45.
- [5] Q. Chen, Z. Zhuo, and W. Wang, “Bert for joint intent classification and slot filling,” *arXiv preprint arXiv:1902.10909*, 2019.